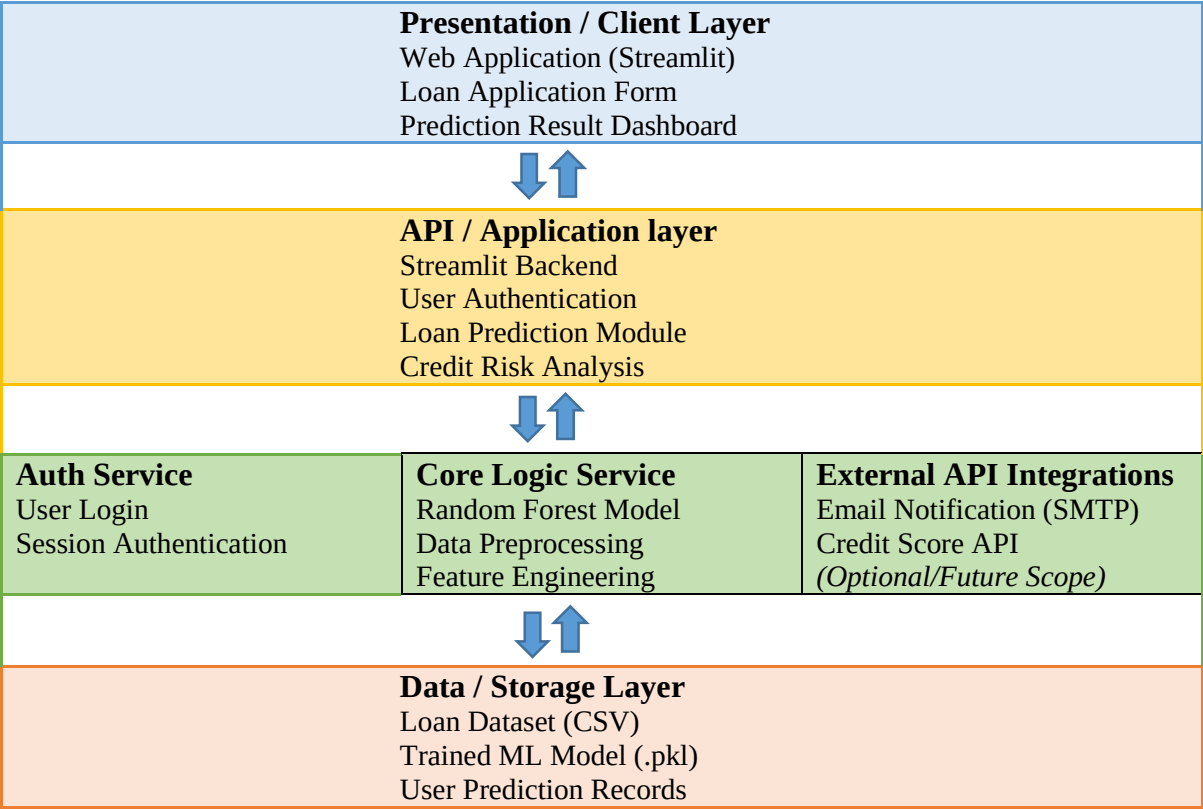


Solution Architecture

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Provides user interface to enter loan details and display prediction results	Streamlit, HTML, CSS
API / Application Layer	Processes user requests, validates inputs, and connects frontend with ML model	Python, Streamlit
Auth Service, Core Logic Service & External API Integrations	Handles user authentication, loan prediction, credit risk analysis, and communication with external notification and credit verification services.	Python, Streamlit, Scikit-learn, Pandas, NumPy, SMTP, Credit Score API (Future Scope)
Data / Storage Layer	Stores the loan dataset, trained machine learning model, and user prediction records required by the application.	CSV Dataset, Pickle (.pkl), Local Storage

